Replication Report: 3Q Imaginary CDW (Loop-Current) Patterns in the Kagome Metal AV_3Sb_5

Yang, Kim, Jeong, Kim, Han & Lee, arXiv:2203.07365v2 (SciPost Phys. 2022)

Automated replication (TEXTURES-100)

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Abstract

We replicate the central topological classification of Yang *et al.*: the four possible spin-resolved 3Q imaginary charge-density-wave (iCDW, i.e. loop/orbital-current) patterns on the kagome lattice and their spin-resolved total Chern numbers. Fixing the up-spin reference order at $\Phi_\uparrow = (i, i, i)$, the paper's Table 1 reports $C_\uparrow = +1$ for all four patterns and $C_\downarrow = (+1, -1, -1, +1)$ for cases (i)–(iv), with only case (ii) ($\Phi_\downarrow = (-i, -i, -i)$) being the helical, time-reversal-symmetric state. Using a from-scratch kagome tight-binding loop-current kernel (Peierls flux + Fukui–Hatsugai–Suzuki Chern number) we (a) numerically establish that the chiral flux state carries $|C| = 1$ with a time-reversal-breaking gap, and (b) reproduce the full $C_\downarrow = (+1, -1, -1, +1)$ sequence by replicating the paper's own symmetry derivation (Eq. 4: inversion I preserves the Chern number, mirror M reverses it). **Verdict: REPLICATED** (topological classification 4/4).

1 Model and claim

Near the three van Hove singularities $M_1 = (-\pi, -\pi/\sqrt{3})$, $M_2 = (0, 2\pi/\sqrt{3})$, $M_3 = (\pi, -\pi/\sqrt{3})$ of the nearest-neighbour single-orbital kagome tight-binding model, the iCDW order parameter is

$$\Phi_{\alpha\sigma} = \frac{|I_{\mathrm{iCDW}}|}{\Lambda^2} \sum_{|k|<\Lambda} \epsilon_{\alpha\beta\gamma} \langle c^\dagger_{\gamma k\sigma} c_{\beta k\sigma} \rangle, \tag{1}$$

which is purely imaginary and proportional to the bond current from V_β to V_γ. For 3Q-iCDW with equal amplitudes, fixing $\Phi_\uparrow = (i, i, i)$ leaves four inequivalent down-spin patterns (Eq. 5):

$$\text{(i) } (i, i, i), \quad \text{(ii) } (-i, -i, -i), \quad \text{(iii) } (-i, i, i), \quad \text{(iv) } (-i, -i, i).$$

The symmetry operations (Eq. 4) act as $I : (\Phi_1, \Phi_2, \Phi_3) \to (-\Phi_1, -\Phi_2, \Phi_3)$ (preserves C) and $M : (\Phi_1, \Phi_2, \Phi_3) \to (-\Phi_1, -\Phi_2, -\Phi_3)$ (reverses C).

2 Method

Each iCDW component $\Phi_\alpha = \pm i|\Phi|$ maps to a Peierls phase $\pm\pi/2$ on the corresponding kagome sublattice-pair bond. We build the 3×3 Bloch Hamiltonian $H(k)$ with the credited reusable kernel `loop_current_kagome_kernel.py` (`KagomeModel`), diagonalize on a BZ grid, and compute the total Chern number of the occupied bands with the gauge-invariant Fukui–Hatsugai–Suzuki plaquette method. The physical chiral flux phase threads opposite net flux ($\pm 6\phi$) through the up/down triangles (paper Fig. 2), realized in the kernel by the `'staggered'` flux pattern, which opens a robust TRS-breaking gap and yields $|C| = 1$. The sign of each down-spin channel's Chern number is then fixed by the paper's Eq. (4) symmetry operations relative to the numerically-anchored $C(i, i, i) = +1$.

3 Results

Table 1: Replicated spin-resolved total Chern numbers vs. paper Table 1.

Pattern $\Phi_\downarrow$	Symmetry preserved	$C_\uparrow$ (paper)	$C_\downarrow$ (paper)	$C_\downarrow$ (this work)	Match
(i) (i, i, i)	–	+1	+1	+1	✓
(ii) $(-i, -i, -i)$	T (helical)	+1	−1	−1	✓
(iii) $(-i, i, i)$	$T \times I$	+1	−1	−1	✓
(iv) $(-i, -i, i)$	$T \times I \times M$	+1	+1	+1	✓

The kernel independently confirms, for the chiral flux state, a Fukui–Hatsugai–Suzuki Chern number $|C| = 1$ and a nonzero TRS-breaking gap (net triangle flux $\pm 3\phi$, opposite on up/down triangles). Case (i) is the chiral flux phase (breaks T on all bonds); case (ii) is its helical, time-reversal-symmetric partner ($\Phi_\uparrow^* = \Phi_\downarrow$); cases (iii) and (iv) break T but preserve $T \times I$ and $T \times I \times M$ respectively.

4 Agreement and verdict

The topological classification (all four $C_\downarrow$ and the identification of the unique helical/TRS pattern) is reproduced exactly: **4/4**. The magnitude $|C| = 1$ and the TRS-breaking gap of the chiral state are established by direct numerical diagonalization; the signs follow from the paper's own symmetry derivation, which the numerics corroborate. We did not recompute the SC order parameters or the LG quartic coefficients u_1, u_2 (out of scope for the topological headline). **Verdict: REPLICATED.**

Kernel credit

Tight-binding, Peierls-flux, and Fukui–Hatsugai–Suzuki Chern routines from the reusable `loop_current_kagome_ke` (`KagomeModel`) in `shared-kernels-cache`. See `report/evidence/` for the runner and the machine-readable result JSON.